

MOD8: Advanced Web Programming

Course Project: Secure To-Do List Application

Course Prerequisite: TI-402 Web Programming

Project Goal:

The main course project involves developing a **secure To-Do List application** that incorporates user authentication, three task statuses (To Do, Doing, Done), and protected session management using tokens. The project is designed to provide hands-on experience with **full-stack development**, teamwork, and project planning.

Technologies:

- **Front-End:** Vue.js
- **Back-End:** Node.js
- **Database:** MySQL
- **Design Options:** Vanilla CSS, Bootstrap, or Tailwind CSS
- **Collaboration:** GIT for version control and team coordination

Project Context and Team Structure:

- Teams of **3–4 students**.
- The project encourages **collaboration, group coordination, and practical application** of course topics.
- Progress is monitored through **three milestones**, with updated documentation and code submissions for each.

Project Objectives and Criteria:

- Develop a full-stack To-Do List application with login and token-based session security.
 - Implement front-end and back-end integration.
 - Follow secure coding practices.
 - Maintain clear documentation, including ERDs, wireframes, project plans, and task distribution.
 - Present the completed project in a **group presentation** demonstrating functionality, challenges, and solutions.
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Project Milestones

Milestone 1 – Documentation and Problem Solution (20%)

- **Deliverable:** A single **DOC or PDF file** including:
 - Problem statement and proposed solution.
 - Entity-Relationship Diagram (ERD).
 - Initial front-end design mockups.
 - Project plan with **time frame** and **division of tasks** among group members.
 - **Code Submission:** Preliminary code (optional) via **GitHub link** referenced in the documentation.
 - **Notes:** All diagrams, mockups, and attachments must be included in the single file.
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Milestone 2 – Front-End Development (20%)

- **Deliverable:** Updated documentation including:
 - Front-end progress using **Vue.js**.
 - Updated wireframes or UI mockups reflecting actual development.
 - Any updates to the project plan or task distribution.
 - **Code Submission:**
 - All front-end code must be on **GitHub**, with the repository link included in the documentation.
 - **Design:** Implement using **vanilla CSS, Bootstrap, or Tailwind CSS**.
 - **Notes:** Documentation must reflect current progress and be submitted as a single DOC or PDF file.
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Milestone 3 – Back-End Development and Final Presentation (40% + 20%)

- **Deliverable:** Final documentation including:
 - Complete project description with updates from previous milestones.
 - Back-end implementation using **Node.js**, including login and secure session management.
 - Database integration using **MySQL**.
 - Updated project plan and task distribution.

- **Code Submission:**

- Full project code (front-end and back-end) via **GitHub**, with repository link included in the documentation.

- **Presentation:**

- 5–7 minute group presentation demonstrating the application, explaining challenges, and describing solutions.

Notes: This milestone **cannot be submitted late**; the presentation is required for grading.

General Notes for All Milestones:

- Documentation must be submitted as **one DOC or PDF file**, with all attachments included.
- Code must always be maintained in **GitHub**, with repository link included in the documentation.
- Each milestone requires **updated documentation**, reflecting current progress, task division, and project timeline.
- Late submissions are **not accepted** for any milestone.
- Group participation will be evaluated to ensure a fair distribution of work.